

GNG 2101
Design Project User and Product Manual

Assistive Pants Device

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List of Acronyms

Table 1. Acronyms

Acronym	Definition
APD	Assistive pants device

1 Introduction

This User and Product Manual (UPM) provides the information necessary for individuals who require assistance in pulling up and down their pants, particularly for users with arthrogryposis or similar conditions that limit their mobility, to effectively use the APD and for prototype documentation. The device enables users to pull their pants up and down independently, promoting dignity, autonomy, and a better quality of life.

We have assumed the following in developing this manual and the device:

1. Users have some basic understanding of operating mechanical devices or have a caregiver who can assist in the setup and initial use.
2. The manual will be accessible to a diverse audience, including individuals with varying levels of literacy or technical expertise.
3. The environment where the device will be used is generally safe, with minimal exposure to extreme temperatures, moisture, or dust.

The document is organized as follows:

- **Introduction:** Explains the purpose of the manual, scope of the device, intended audience, and safety/privacy considerations.
- **Overview:** Summarizes the problem addressed, user needs, key features, and device construction in non-technical terms. Includes a block diagram and images of the prototype.
- **Getting Started:** Walks the user through setup, configuration, and access, with visual aids to make the process clear and intuitive.
- **Using the System:** Details the functions and features of the device with step-by-step instructions and accompanying visuals.
- **Troubleshooting and Support:** Provides solutions for common errors, maintenance tips, and contact information for support.

- **Product Documentation:** Describes the design, materials, calculations, and construction methods. Includes a bill of materials (BOM), equipment lists, and instructions for recreating the device, supported by diagrams and photos.
- **Testing and Validation:** Presents testing methods and results, highlighting the device's performance and limitations.

Purpose, Scope, and Audience

- **Purpose:** The document serves as a comprehensive guide to ensure the safe and effective use of the assistive pants device.
- **Scope:** Covers all aspects of the device lifecycle, including setup, operation, maintenance, and disposal. It does not cover advanced repairs requiring specialized tools or expertise.
- **Intended Audience:** End users (individuals with limited mobility), caregivers, and healthcare professionals who may assist in its usage.

Security, Safety, and Privacy Considerations

- **Security:** Users should store the manual securely to avoid misplacing critical operational instructions.
- **Safety:** Clear warnings are provided to prevent misuse, such as overextending the telescopic rods, improper handling of magnets, or overinflating the air pump mechanism. The manual emphasizes ensuring the device is free of damage before use.
- **Privacy:** The manual advises users to handle the device in private areas to maintain personal dignity. Additionally, no user data is collected or stored by the device.

This document aims to empower users with clear and accessible guidance while maintaining a focus on safety, usability, and respect for individual privacy.

2 Overview

For individuals with limited mobility, particularly those with arthrogryposis or similar conditions, daily tasks such as dressing can be challenging, time-consuming, and reliant on caregiver assistance. This dependency can affect their independence, privacy, and overall quality of life. Addressing this issue is critical to promote autonomy and dignity for individuals who face these challenges.

User needs:

1. Must be able to pull over butt easily
2. Can operate device independently
3. Have hands in front of body to operate device
4. Uses minimal shoulder and elbow movement
5. Device is portable (on body or wheelchair to be determined)
6. Device is adaptable for size (grows with client)

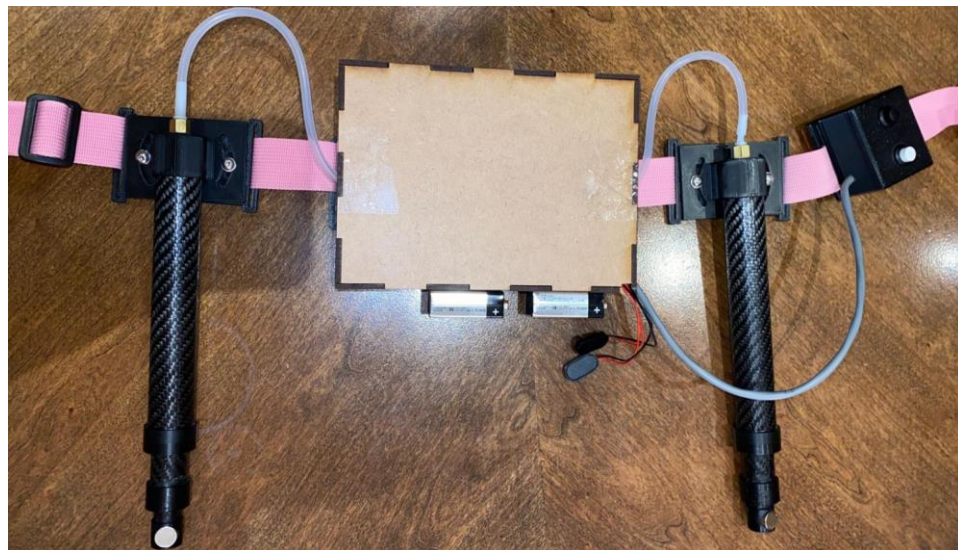


Figure 1. Final APD Prototype

Differentiating Features:

- **Magnetic System:** Provides secure but easily adjustable attachments to the user's clothing, ensuring smooth operation.
- **Carbon Telescopic Rods:** Lightweight yet durable construction allows the device to be easily transported and used in different settings.
- **Air Pump Mechanism:** Reduces the effort needed to lift and lower pants, offering seamless motion with minimal user exertion.
- **User-Centric Design:** Designed with input from users with limited mobility to ensure it meets their specific needs effectively.
- **Compact and Foldable Design:** Ensures portability and convenient storage.

Architecture/Construction:

- **Frame:** Carbon telescopic rods form the base, ensuring strength and portability.
- **Mechanism:** Operates through a pneumatic air pump system for smooth lifting and lowering actions.
- **Attachments:** Magnets connect to the pants, making setup easy and quick.
- **User Access Mode:** The device is activated through a straightforward button, requiring minimal strength or dexterity.

Cautions & Warnings

Operational Warnings:

- **Warning:** Do not overinflate the air pump as this can damage the telescopic rods or the pneumatic system.
- **Warning:** Keep fingers clear of moving parts during operation to avoid pinching or injury.

Environmental Considerations:

- **Caution:** Avoid exposing the device to extreme temperatures, moisture, or dust, as these conditions may affect performance.

Maintenance Warnings:

- **Warning:** Ensure all components are securely fastened and free of visible damage before use.

User-Specific Warnings:

- **Caution:** Individuals with limited hand strength or dexterity should seek assistance with initial setup.
- **Warning:** The device is not intended to lift heavy loads or be used for purposes other than its designed function.

3 Getting started

Setup Instructions

1. Remove the device from its packaging and check for all components
 - Telescopic rods
 - Magnets
 - Central electronics unit
 - Straps and accessories
2. Secure the adjustable belt around the waist.
 - Ensure the belt is snug but not overly tight.



Figure 2. Attaching Belt Around Waist

3. Connect Magnetic Tethers to Clothing

- Attach the magnets to the waistband of the pants.
- Make sure the magnets are securely fixed and centered.



Figure 3. Connecting Magnets to Pants

Operation Instructions

4. Activate the Device

- Press the upper black button to extend the rods
- Press the lower white button to retract the rods
- Observe the telescopic rods as they extend or retract, ensuring smooth operation



Figure 4. Extending Rods



Figure 5. Retracting Rods

5. Pause or Stop Operation

- To pause, release the button.
- For complete shutdown, disconnect battery.

Exit Instructions

6. Detach Magnetic Tethers

- Gently remove the magnets from the pants.



Figure 6. Removing Belt

7. Store the Device

- Store all components in a safe location.
- Ensure the tubing is not kinked or damaged.

Configuration Considerations

Physical Prototype

The APD includes the following components:

- Telescopic Rods: Carbon fiber rods that extend/retract to lift or lower pants.
- Magnets: Attach securely to the waistband of pants.
- Control Unit: Compact box housing the air pump, vacuum, and other hardware.
- Waist Strap: Adjustable strap securing the device around the user's waist.
- Air Tubing: Flexible tubes connecting the control unit to the telescopic rods.

Setup and Tools

- Tools Required: None.
- Connections: Attach magnets to pants and secure waist strap.

4 Using the System

4.1.1 Using the System

This section provides a detailed guide for each function and feature of the Assistive Pants Device (APD). Users can refer to the following sub-sections to understand how to interact with the device effectively. Each feature is explained with clear instructions and visuals where necessary.

4.1.1.1 4.1 Device Activation

Input Required: Press the designated buttons to activate or deactivate the telescopic rods.

Output Produced: The rods will extend or retract based on the button pressed.

Steps:

1. Power On the Device:

2. Ensure the batteries are securely connected to the control unit. The power indicator should light up to confirm the device is ready for use.

3. Extending the Rods:

- a. Press the **upper black button**.
- b. The air pump will activate, inflating the telescopic rods to lift the pants.
- c. Observe the smooth extension of the rods, ensuring there is no obstruction in their path.

4. Retracting the Rods:

- a. Press the **lower white button**.
- b. The vacuum system will engage, pulling air out of the rods to collapse them.
- c. Ensure the rods fully retract for proper storage and operation.

Cautions:

- Avoid pressing both buttons simultaneously.
- If the rods do not respond, check the battery connection and air tubing.

4.1.1.2 4.2 Adjusting the Fit

Input Required: Adjust the waist strap and ensure magnets are properly aligned.

Output Produced: The device fits securely on the user and connects efficiently to their pants.

Steps:**1. Securing the Waist Strap:**

- a. Wrap the adjustable strap around the waist, ensuring it is snug but comfortable.
- b. Fasten the buckle and make necessary adjustments using the strap's slider.

2. Attaching Magnets to Clothing:

- a. Secure the magnetic tethers to the waistband of the pants.
- b. Align the magnets evenly on both sides for balanced operation.
- c. Double-check that the magnets are firmly attached.

Cautions:

- Ensure the belt is not too tight, as this may cause discomfort.
- Check that magnets are securely in place before activating the device.

4.1.1.3 4.3 Pause and Shutdown Operations

Input Required: Release the buttons to pause operation or disconnect the batteries for a complete shutdown.

Output Produced: The system halts operation or powers down.

Steps:

1. Pausing the Device:

- a. Release the button mid-operation to stop the rods from extending or retracting further.
- b. The rods will remain in their current position until a button is pressed again.

2. Shutting Down the Device:

- a. Disconnect the battery pack from the control unit when the device is not in use.
- b. Store the batteries and the device in a safe, dry location.

Cautions:

- Always ensure the rods are fully retracted before disconnecting the power.
- Avoid storing the device with the battery connected to prevent accidental activation.

4.1.1.4 4.4 Maintenance and Storage

Input Required: Regular inspection of device components and proper storage practices.

Output Produced: Ensures longevity and optimal performance of the device.

Steps:

1. Inspecting the Device:

- a. Check the rods for signs of wear or damage.
- b. Ensure the air tubing is not kinked or punctured.

2. Cleaning:

- a. Use a damp cloth to clean the control unit and telescopic rods. Avoid submerging any part of the device in water.
- b. Remove dust or debris from the magnets.

3. Storage:

- a. Store the device in a dry, safe place, ensuring no pressure is applied to the rods or tubing.
- b. Keep the manual and all accessories together for easy access.

Cautions:

- Do not use abrasive cleaning materials.
- Avoid exposure to extreme temperatures during storage.

5 Troubleshooting & Support

This section provides clear instructions for resolving common issues and maintaining the Assistive Pants Device (APD). It also outlines the support options available for users in case of persistent problems.

Error Messages or Behaviors

Below are common issues, their potential causes, and recommended solutions:

- **Device does not power on**
 - *Possible Cause:* Dead or improperly connected batteries
 - *Corrective Action:* Replace or reconnect the 9V batteries; ensure polarity matches the markings.
- **Rods fail to extend or retract**
 - *Possible Cause:* Air tubing is kinked or disconnected
 - *Corrective Action:* Inspect tubing for kinks or detachment; reattach securely and ensure clear airflow.
- **Weak or incomplete rod movement**
 - *Possible Cause:* Low battery power or air leakage in seals
 - *Corrective Action:* Replace batteries; inspect seals for wear or gaps, and reseal the O-rings if needed.
- **Magnets do not attach securely**
 - *Possible Cause:* Misaligned or weak magnetic tethers
 - *Corrective Action:* Realign magnets; clean both the magnets and the pants' surface to remove debris.
- **Buttons unresponsive**
 - *Possible Cause:* Loose wiring or mechanical failure
 - *Corrective Action:* Check connections inside the control unit; if unresolved, contact support.
- **Rods retract but not fully collapse**
 - *Possible Cause:* Debris or obstruction in the telescopic rods

- *Corrective Action:* Clean rods with a damp cloth and ensure they are free of obstructions.

Special Considerations

Certain scenarios require extra attention to ensure proper device operation:

1. Extreme Environments:

- a. Avoid using the device in areas with excessive dust, moisture, or extreme temperatures, as these can affect the air pump and electronic components.

2. User Strength and Dexterity:

- a. Users with severely limited strength may need caregiver assistance for initial setup, particularly when attaching magnets or securing the waist strap.

3. Battery Handling:

- a. Use only the recommended 9V batteries. Do not mix old and new batteries, as this can reduce performance. Dispose of depleted batteries responsibly.

4. Frequent Disassembly:

- a. Repeated disassembly of components (e.g., magnets or tubing) may increase wear. Perform regular inspections if frequent disassembly is unavoidable.

Maintenance

Regular maintenance ensures the APD remains in optimal condition. Perform the following tasks at least once every two weeks or as needed:

1. Inspect the Device:

- a. Check the telescopic rods, seals, and air tubing for visible wear or damage. Replace worn components promptly.

2. Clean the Device:

- a. Wipe all parts with a damp cloth to remove dust and debris. Avoid using abrasive cleaners or submerging parts in water.

3. Lubricate Moving Parts:

- a. Apply a small amount of silicone lubricant to the telescopic rods to ensure smooth movement.
- 4. Battery Replacement:**
 - a. Replace the 9V batteries when the device exhibits slower movement or reduced power.
- 5. Check for Air Leaks:**
 - a. Test the seals by activating the air pump while holding the rods in place. If you hear hissing or detect a leak, inspect the O-rings and replace them if necessary.

Support

This device is not a threat to human life if there is a malfunction, if you have any concerns or issues with the device please contact:

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6 Product Documentation

6.1 Subsystems of Prototypes

Prototype 1

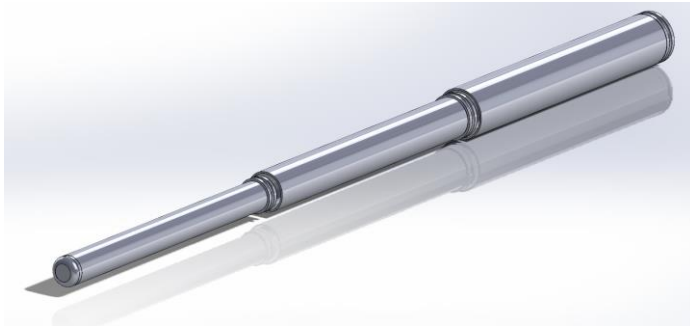


Figure 7. CAD model of telescopic arm extension

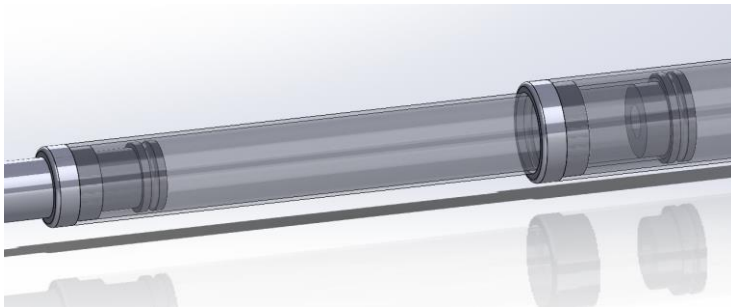


Figure 8. Airtight male and female valve connectors

Prototype 2



Figure 9. Completed carbon rods to required dimensions



Figure 10. Male/female airtight valve version 1

6.1.1 BOM (Bill of Materials)

Item Number	Element	Material	Quantity	Specs	Total Cost
Structural Frame					
	Telescopic cylinder bottom	carbon	3	prepreg 200gsm carbon	30
	Male/femal ends	PLA	6	3D print	1
	Telescopic rod connector to belt	PLA	4	3D print	0.3
	Magnets	Magnets	2		1
	O-Ring	Buna-n	6	14.5 and 18mm	10
Air Pump System					
	Value	Various	2	6V with 2-pin JST XH Connector	10
	Pump/Vacuum	Various	4	DC motor - 4.5V and 1.8LPM	35
	PWM Motor Driver	Various	5	5V DC - 15A	12
	Hose	silicon	2	200mm	8
Control System					
	Wire	copper	2m		0.5
	Arduino Uno	various	1		15
	Buttons	Various	1		5
Belt					
	Nylon Belt and buckle	fabric	1		5
	3D printed arduino/pump box	pla	1	3D print	3
					135.8

6.1.2 Equipment list

- Carbon Fiber Rods (pre-cut for telescopic mechanism)
- O-Rings (for sealing)
- End Caps (designed for rod alignment and magnetic attachment)
- Neodymium Magnets (for secure attachment to clothing)
- Silicone Lubricant (for smooth operation of the rods)
- ABS Plastic Enclosure (housing control mechanisms)

6.1.3 Instructions

Step 1: Preparing the Telescopic Rods

1. Gather carbon fiber rods of varying diameters to allow smooth telescoping.
2. Attach O-rings to the smaller-diameter ends of the rods for airtight sealing.
3. Secure end caps at the ends of each rod segment, ensuring they align properly for smooth extension and retraction.

Step 2: Installing Magnets

1. Embed neodymium magnets into protective plastic casings to prevent damage.
2. Fix the casings securely onto the rod end caps.
3. Test the magnetic force on sample clothing to ensure a secure hold.

Step 3: Assembling the Control Unit Housing

1. Use the ABS plastic box to house the pneumatic system and rods when collapsed.
2. Create slots for the air tubing and button connections, ensuring they align with the internal components.

Testing & Validation

Durability Testing:

- Simulated over 10,000 cycles of extension and retraction to validate the lifespan of the rods and O-rings.
- Checked for consistent sealing and rod alignment under repeated use.

Performance Testing:

- Timed rod extension and retraction cycles to ensure smooth operation.
- Measured the holding force of magnets to confirm secure attachment during simulated use cases.

User Testing:

- Conducted trials with individuals with limited mobility to gather feedback on ease of use and setup.
- Adjusted rod length and magnet placement based on user preferences.

Issues Encountered:

- Initial O-ring designs leaked air during testing; replaced with higher-grade sealing materials.
- Early prototypes used ferrite magnets, which proved insufficient for reliable attachment

7 Conclusions and Recommendations for Future Work

The development of the Assistive Pants Device (APD) has been a valuable learning experience, emphasizing the importance of iterative prototyping, user feedback, and interdisciplinary collaboration. While the current design achieves its primary objective of enhancing independence for individuals with limited mobility, there are several opportunities for further improvement and refinement.

7.1.1.1 Lessons Learned

1. **User-Centered Design Drives Success:** Engaging with users early and frequently helped shape features like button-based controls and portable construction. Their feedback was essential in addressing real-world challenges.
2. **Material Selection Matters:** The choice of carbon fiber for the telescopic rods proved effective in balancing weight, strength, and durability. However, achieving airtight seals with O-rings required additional testing and adjustments.
3. **Testing Protocols Need Refinement:** While theoretical calculations supported the device's performance, real-world testing uncovered practical issues, such as air leaks and magnet misalignment, that required rapid iteration.

7.1.1.2 What Could Be Achieved with More Time

If given additional time, the following enhancements would be prioritized:

1. **Improved Sealing Mechanisms**
 - a. Refining the O-ring design or exploring alternative sealing technologies to enhance air retention and rod performance.
 - b. Conducting extended durability testing to validate long-term performance in diverse environments.
2. **Optimizing Air Pump Efficiency**

- a. Replacing the current pump with a more powerful and quieter model to reduce operation time and improve user experience.
 - b. Exploring alternative actuation systems, such as motorized extensions, to simplify maintenance.
- 3. Enhanced Portability and Adaptability**
- a. Miniaturizing the control unit for increased portability and discreetness.
 - b. Incorporating adjustable components to better accommodate diverse user needs and clothing styles.
- 4. Advanced Testing and Validation**
- a. Establishing a comprehensive test plan covering usability, durability, and safety under various conditions.
 - b. Including a larger user group in trials to gather broader feedback and refine the device accordingly.
- 5. Cost Reduction and Accessibility**
- a. Exploring cost-effective materials and manufacturing processes to make the device more affordable without compromising quality.
 - b. Simplifying the design to reduce assembly time and improve scalability for mass production.

7.1.1.3 Abandoned Ideas Due to Time Constraints

- 1. Wireless or App-Based Controls:**
 - a. Integrating wireless technology for remote operation could benefit caregivers and users with severe mobility limitations. This feature was set aside to focus on core functionality.
- 2. Enhanced User Feedback Systems:**
 - a. Tactile or audio feedback for button presses was considered but not implemented due to time constraints.

3. Advanced Magnet Attachments:

- a. Developing interchangeable attachment options for various clothing types could expand the device's versatility.

By addressing these areas in future iterations, the APD can evolve into an even more effective and accessible solution for individuals with limited mobility. The foundation laid in this project provides a solid starting point for subsequent teams to innovate further, empowering more users and improving their quality of life.

8 Appendix

8.1 Design Files

Table 2. Referenced Documents

Document Name	Document Location and/or URL	Issuance Date
Video_Manual_BA1	https://drive.google.com/file/d/1OxxSCx4yeJ63lVuVx8YT45EF6RX2Oni8/view?usp=sharing	3/12/24